

III MATTER

Deformation of solids
AS

AS

Ideal gases
Temperature
Thermodynamics

A2

$$pV = N k T$$

$$n = \frac{N}{N_A}$$

Pressure of gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

Average kinetic energy
per molecule

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

$$pV = nRT$$

10.1
Equation of
State

10.2 Kinetic
theory of
gases

10.3
Pressure
of a gas

10.4 Kinetic
energy of a
molecule

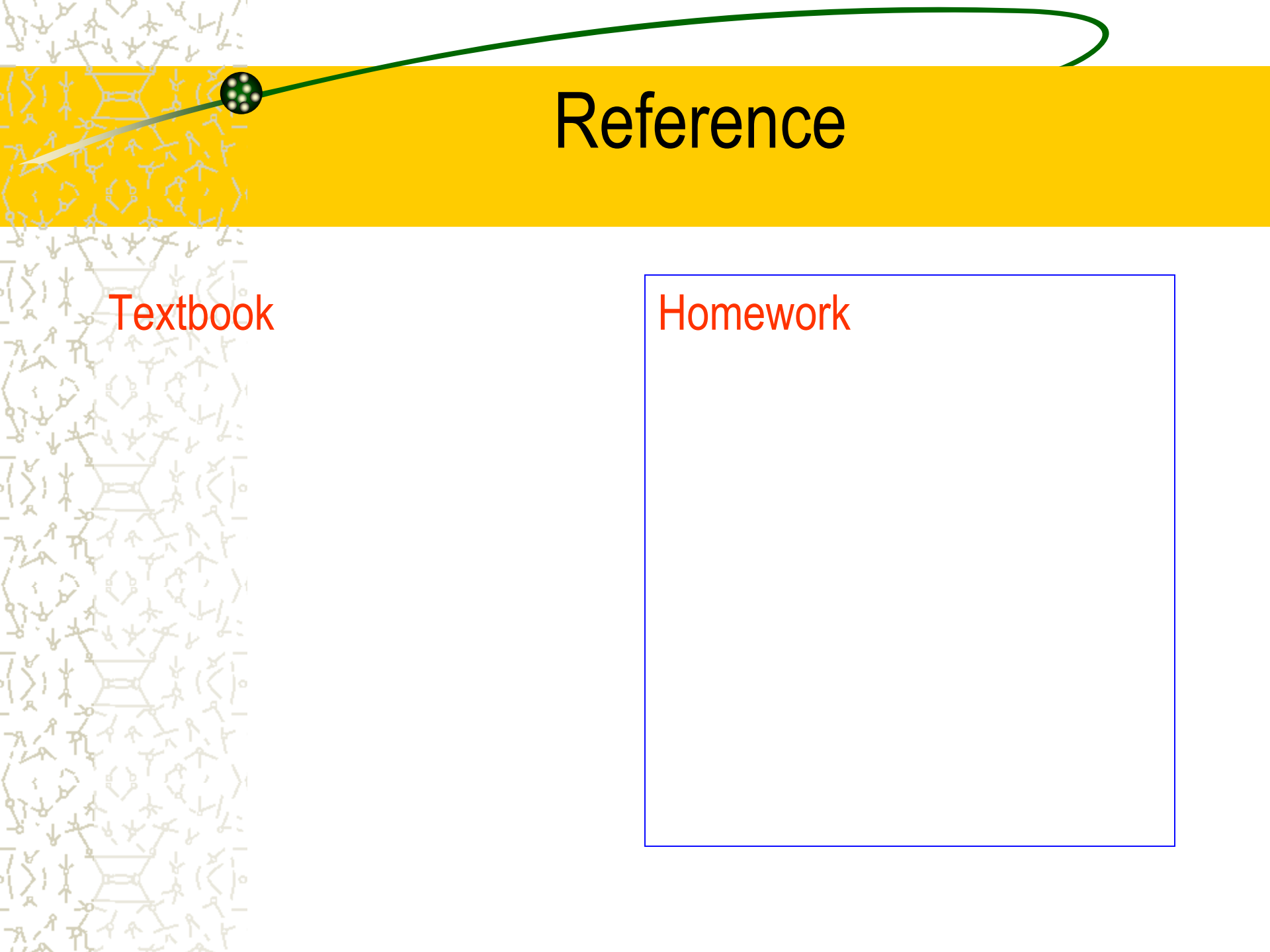
10. IDEAL GASES

- 11.1 Equation of state
- 11.2 Kinetic theory of gases
- 11.3 Pressure of a gas
- 11.4 Kinetic energy of a molecule

Learning Outcomes

Candidates should be able to:

- (a) recall and solve problems using the equation of state for an ideal gas expressed as $pV = nRT$. (n = number of moles)
- (b) infer from a Brownian motion experiment the evidence for the movement of molecules.
- (c) state the basic assumptions of the kinetic theory of gases.
- (d) explain how molecular movement causes the pressure exerted by a gas and hence deduce the relationship, $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$. (N = number of molecules)
[A rigorous derivation is not required.]
- (e) compare $pV = \frac{1}{3} Nm \langle c^2 \rangle$ with $pV = NkT$ and hence deduce that the average translational kinetic energy of a molecule is proportional to T .



Reference

Textbook

Homework

THE GAS LAWS

How do gases behave if their **pressure, volume** or **temperature** is changed.

It is sensible to **vary two** of the previous quantities while keeping the other constant in three separate experiments:

- (i) Variation of **pressure with volume** at a constant temperature
- (ii) Variation of **pressure with temperature** at a constant volume
- (iii) Variation of **volume with temperature** at a constant pressure

Parameters

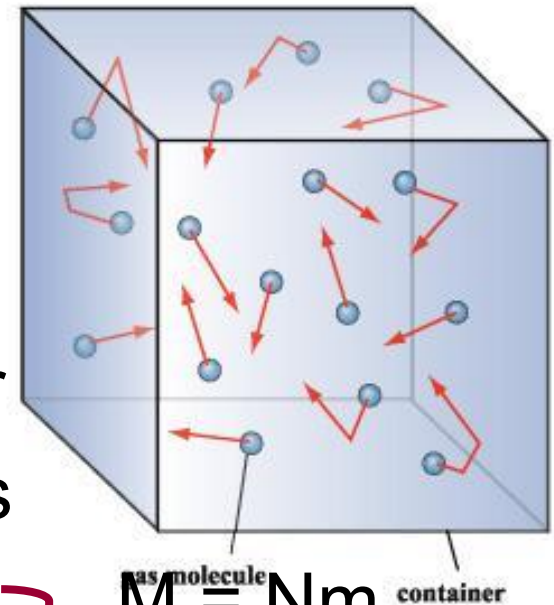
V - volume of container

p - pressure of gas in container

T - absolute temperature of gas

N - number of molecules of gas

m = mass of a gas molecule

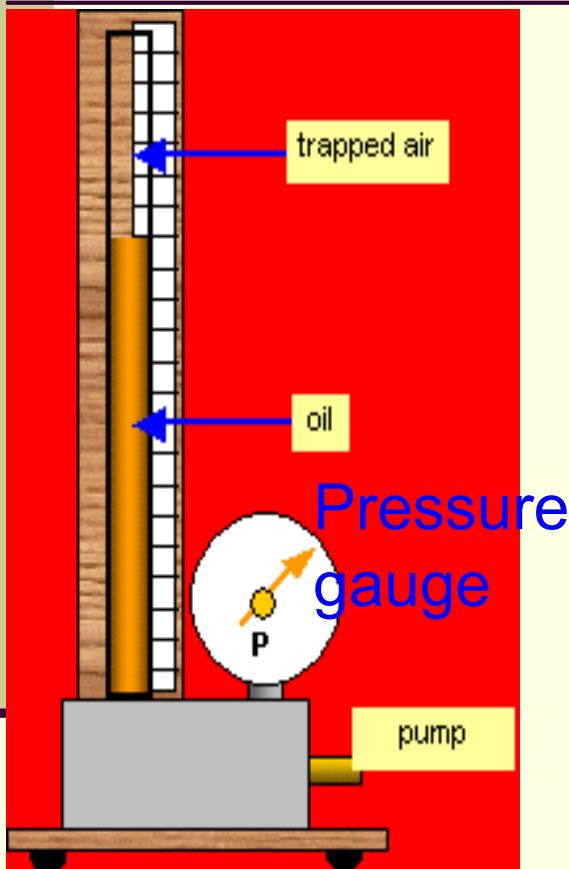


$$M = Nm$$

total mass
of gas

V, p and T are called **macroscopic** properties
(what we can see and measure).

VARIATION OF PRESSURE WITH VOLUME



This can be investigated using the apparatus shown in the diagram. The air trapped in the glass tube is compressed by forcing in oil with the pump and taking readings of pressure and volume. After each compression you should wait a few moments to allow the temperature of the air to stabilise.

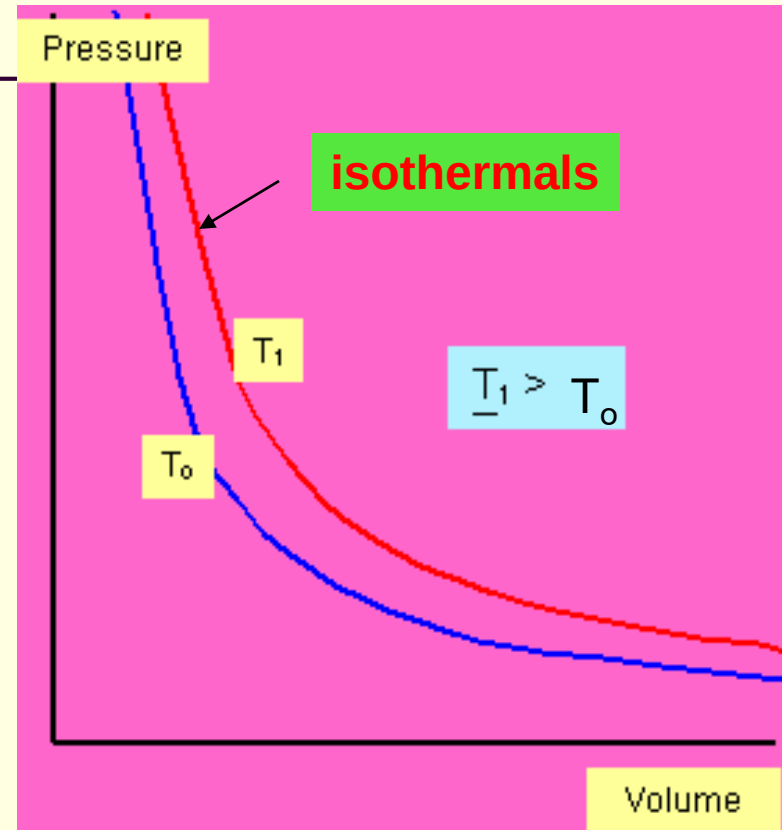
The relation between pressure and volume was first discovered by Robert Boyle in 1660 and is called **Boyle's Law**. It states that:

$$pV = \text{constant}$$

Boyle's Law

A graph of pressure against volume is shown in the following diagram for two different temperatures T_1 and T_0 ($T_1 > T_0$). The lines on it are **isothermals**, that is they join points of equal temperature.

If a fixed mass of gas with a pressure P_1 and a volume V_1 changes at **constant temperature** to a pressure P_2 and volume V_2 Boyle's Law can be written as:



$$P_1 V_1 = P_2 V_2$$

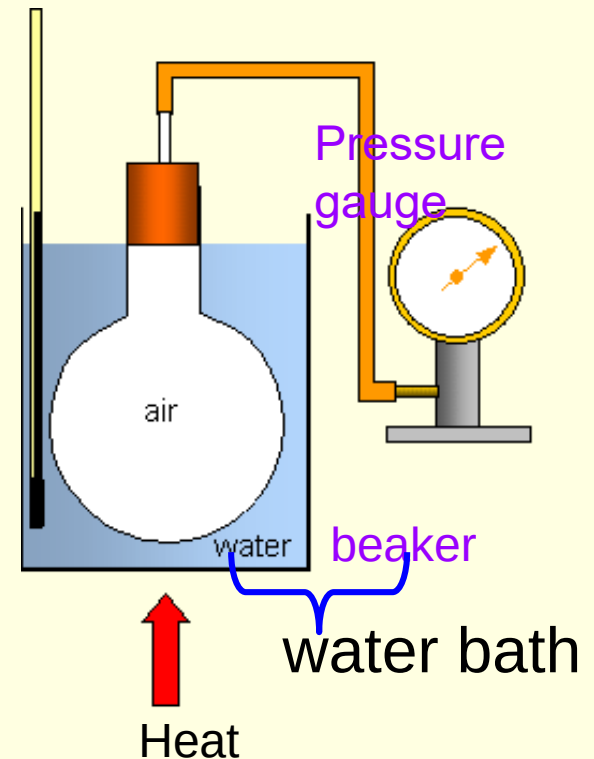
VARIATION OF PRESSURE WITH TEMPERATURE (Pressure Law)

The water is heated and the pressure of the air in the sealed glass beaker is measured with the pressure gauge. (The volume of the air is effectively constant).

Results of this experiment show that for a fixed mass of gas at constant volume:

V and M constant

$$\text{Pressure (P)} = \text{constant} \times \text{absolute temperature (T)}$$



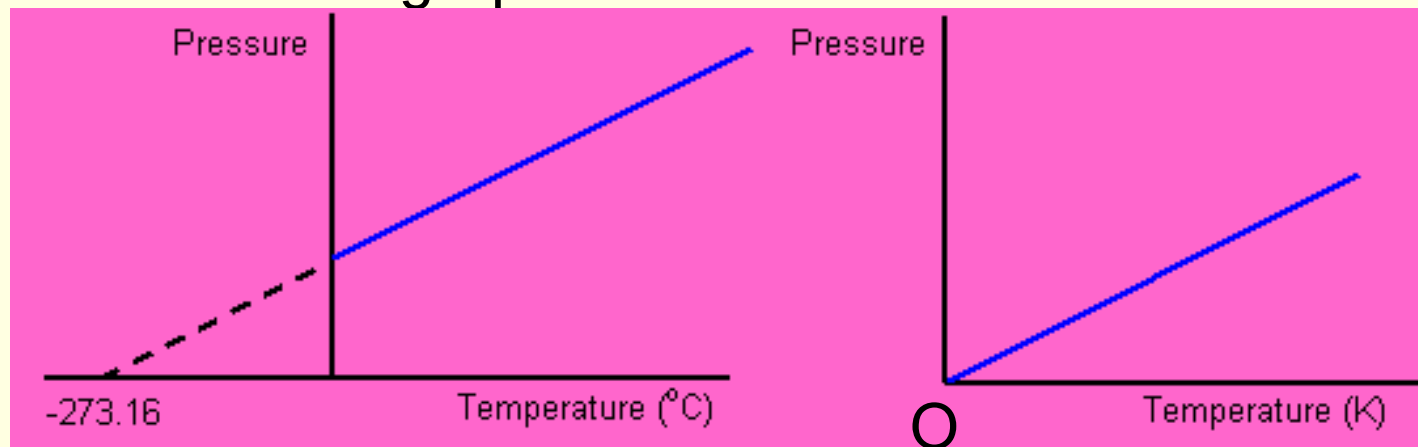
Constant volume
gas thermometer

PRESSURE LAW

If a fixed mass of gas with a pressure P_1 and a temperature T_1 changes to a pressure P_2 and temperature T_2 with no change of volume this can be written as:

$$\text{Pressure law for a gas: } P_1 / T_1 = P_2 / T_2$$

The variation of the pressure of the air with temperature is shown in the graphs below.



CHARLES' LAW

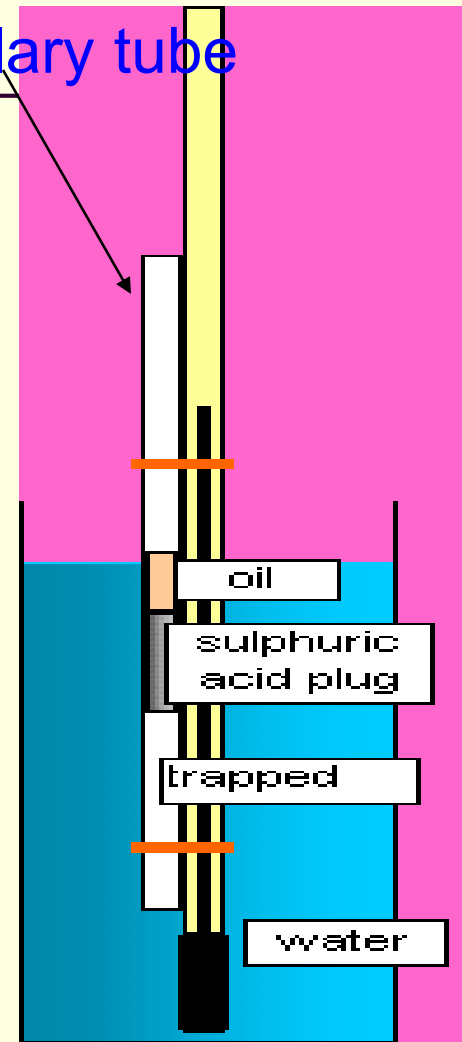
- The capillary tube has a small plug of concentrated sulphuric acid placed in it and it is then sealed at the other end.

(It is most important that appropriate safety precautions are taken when carrying out this experiment. Your eyes must be protected.)

- The water in the beaker is heated and the length of the trapped air column and the temperature are both recorded. Results of this experiment show that for a fixed mass of gas at constant pressure.

$$\text{Volume}(V) = \text{constant} \times \text{absolute temperature } (T)$$

Capillary tube



VARIATION OF VOLUME WITH TEMPERATURE

If a fixed mass of gas with a volume V_1 and a temperature T_1 changes to a volume V_2 and temperature T_2 with no change of volume this can be written:
p and M constant

$$\begin{aligned} &\text{Initial volume/Initial temperature} \\ &= \text{Final volume/Final temperature} \end{aligned}$$

or

$$V_1/T_1 = V_2/T_2$$

Gas

- When working with gases we preferred to work with a quantity called the number of mole rather than mass of gas.
- Absolute temperature: $T/K = \theta/^{\circ}\text{C} + 273$

Definition

- **1 mole** (or mol) is the amount of substance, which contains as many elementary units or entities as there are atoms in 12g of ^{12}C .

e.g. 1 mole = 2 g of H_2
= 32 g of oxygen gas.

- (entities may be atoms, molecules, ions, electrons or other particles).

Avogadro constant (N_A) is the number of atoms in 0.012 kg of carbon-12.

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

- If there are N molecules in a container, then the number of mole of the substance is

$$n = \frac{N}{N_A}$$

Molar mass

- The ***molar mass*** (M_r) is defined as the mass of one mole of the substance
- unit : g mol^{-1}
- e.g. molecular mass of $^{12}\text{C} = 12 \text{ g mol}^{-1}$

For M kg of a substance of molar mass M_r , the number of mole,

$$n = \frac{M}{M_r}$$

Example 11.1

$$n = \frac{N}{N_A}$$

12g of carbon-12 contains
 6.02×10^{23} atoms.

Calculate (a) the mass of one carbon - 12 atom and
(b) the average mass of a nucleon (This is the atomic mass unit). (A nucleon is a particle found in the nucleus namely proton or neutron).

(Ans. 1.99×10^{-26} kg, 1.66×10^{-27} kg)

a) 6.02×10^{23} atoms has a
mass of 12 g

mass of one atom

$$= \frac{12 \text{ g}}{6.02 \times 10^{23}} = 1.99 \times 10^{-26} \text{ kg}$$

b) there are 12 nucleons in
the nucleus.

mass of nucleon

$$= \frac{1.99 \times 10^{-26}}{12} = 1.66 \times 10^{-27} \text{ kg}$$

Example 11.2

Calculate a) the number of atoms in 0.3 g of lithium (${}^7\text{Li}$), and b) the number of moles of lithium. (Ans. 2.58×10^{22} atoms, 0.043 mole)

$$n = \frac{M}{M_r}$$

a) 7 g of Lithium contains 6.02×10^{23} atoms.

$$\begin{aligned}\therefore 0.3 \text{ g contains} \\ &= \frac{6.02 \times 10^{23} (0.3)}{7} \\ &= 2.58 \times 10^{22} \text{ atoms}\end{aligned}$$

b) no. of moles = $0.3/7$
 $= 0.043$

Equation of state (Ideal gas equation)

$$pV = nRT$$

Combining the equations

$PV = \text{constant}$,

$P/T = \text{constant}$ and

$V/T = \text{constant}$ gives:

$$\frac{pV}{T} = \text{constant}$$

For 1 mole of gas the constant is known as the **molar gas constant (R)**

$$pV = RT$$

Now the volume of **one mole** of an ideal gas at **Standard Temperature and Pressure (STP)** (1.014×10^5 Pa and 273.15 K) is 0.0224 m^3 and so

$$1.014 \times 10^5 \times 0.0224 = 1 \times R \times 273.15 \text{ and therefore}$$

$$R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}.$$

[given in "DATA SHEET"]

Ideal gas equation (alternative)

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

$$pV = N k T$$

- k = Boltzmann's constant =
 R/N_A
 $= 8.31/6.02 \times 10^{23}$
 $= 1.38 \times 10^{-23} \text{ J K}^{-1}$

- N = number of molecules

[Values given in "DATA SHEET"]

Derivation

$$pV = nRT$$

$$pV = \frac{N}{N_A} RT$$

$$= pV = N k T$$

What is an **ideal gas**?

An **ideal gas** is one that obeys the gas laws, and equation of state for ideal gas, at all temperature, pressure and volume.

Examples are oxygen and nitrogen near room temperature, carbon dioxide gas can be liquefy near room temperature, thus does not obey Boyle's law.

Many gases at room temperature and moderate pressure behave as ideal gas.

What is an **ideal gas**?

- The **internal energy** (U) is entirely kinetic energy, and depends on its absolute temperature. $U = \frac{3}{2} NkT$
- The behaviour of real gas (and unsaturated vapour) can be described by $pV = nRT$ if they are at low temperature which are well above those at which they liquefy.

Example 11.3

A volume 250 cm^3 of gas is trapped in a cylinder closed by a smooth piston, at a pressure of $1.2 \times 10^5 \text{ Pa}$. The piston is pushed in slowly until the volume of gas is 150 cm^3 , what is the new pressure. (Ans. $2.0 \times 10^5 \text{ Pa}$)

Solution

Pushed in slowly means the temperature is constant.

$$p_1 V_1 = p_2 V_2$$

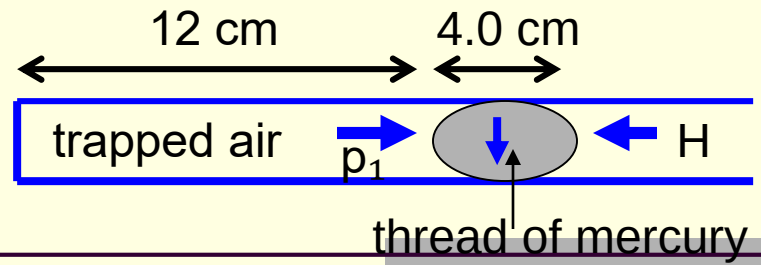
$$1.2 \times 10^5 (250) = p_2 150$$

$$p_2 = 2.0 \times 10^5 \text{ Pa}$$

Example 11.4

A uniform capillary tube is closed at one end by a thread of mercury of length 4.0 cm. When the tube is placed horizontally the column of air has a length of 12 cm. Take the atmospheric pressure to be 76 cmHg. The tube has a cross-sectional area of 20 cm^2 .

a) What is the pressure of the trapped air?



Solution

H = atmospheric pressure
= 76 cmHg

$A = 20 \text{ cm}^2$

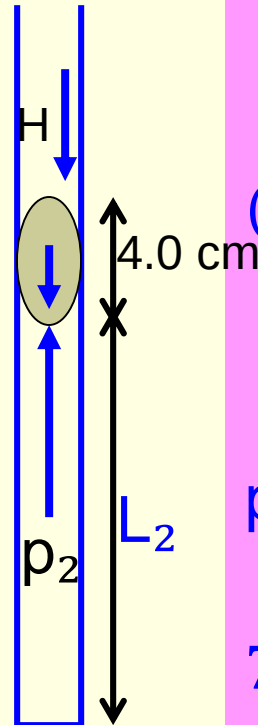
$V_1 = 12A \text{ cm}^3$

a) p_1 = pressure of trapped air = 76 cmHg
(p_1 to the right equals the atmospheric pressure to the left)

Example 11.4

- b) When the tube is held vertically,
 i) with the open end upwards, what is the length of the column of trapped air?

A = cross-sectional area of tube



$$V_2 = AL_2$$

$$p_2 = (H + 4) = 80 \text{ cmHg}$$

(p_2 supports the mercury thread and atmospheric pressure)

$$p_1 V_1 = p_2 V_2$$

$$76(12A) = 80AL_2$$

$$L_2 = 11.4 \text{ cm}$$

Example 11.4

ii) With open end downwards?

$$p_3 + 4 = H$$

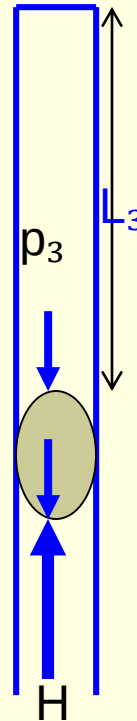
$$p_3 = 76 - 4 = 72 \text{ cmHg}$$

(atmospheric pressure supports the mercury thread and trapped gas)

$$p_1 V_1 = p_3 V_3$$

$$76(12A) = 72(A L_3)$$

$$L_3 = 12.7 \text{ cm}$$



c) If the temperature of the gas is 27°C, calculate the number of mole of gas enclosed.

$$\begin{aligned} n &= \frac{pV}{RT} \\ &= \frac{1.01 \times 10^5 (0.12) 20(10^{-4})}{8.31(273 + 27)} \\ &= 9.98 \times 10^{-3} \text{ mol.} \end{aligned}$$

Example 11.5

A mass of carbon dioxide occupies 15.00 m^3 at 10°C and 101.97 kPa .

a) What will be its volume at 40.0°C and 106.63 kPa ?

Calculate

b) the number of mole of gas,

c) the number of molecules of gas and

d) the mass of gas if the molar mass of CO_2 is 44 g .

(Ans. a) 15.9 m^3 , b) 651 ,
c) 3.95×10^{26} molecules,
d) $2.86 \times 10^4 \text{ g}$)

Solution

$$\text{a) } \frac{p_2 V_2}{T_2} = \frac{p_1 V_1}{T_1}$$

$$\frac{106.63 V_2}{(273 + 40)} = \frac{101.97(15)}{(273 + 10)}$$

$$V_2 = 15.9 \text{ m}^3$$

$$\begin{aligned} \text{b) } n &= \frac{pV}{RT} \\ &= \frac{101.97 \times 10^3 (15)}{8.31(273 + 10)} \end{aligned}$$

$$= 651 \text{ moles}$$

Example 11.5

$$n = \frac{N}{N_A}$$

$$n = \frac{M}{M_r}$$

c) 1 moles contains 6.02×10^{23} molecules

$\therefore$ 651 moles contains

$$651 \times 6.02 \times 10^{23} = 3.92 \times 10^{26} \text{ molecules}$$

d) mass of gas

$$= 651(44) = 2.86 \times 10^4 \text{ g}$$

Example 11.6

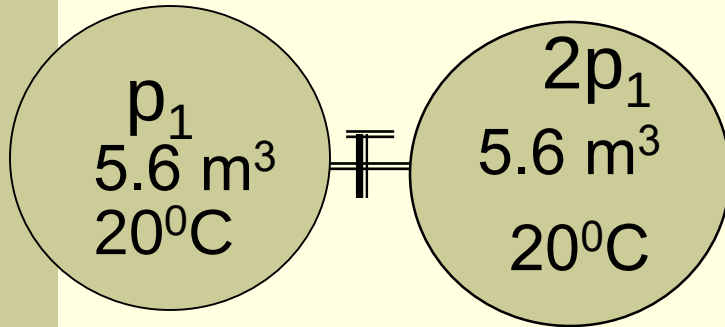
Two flasks having equal volumes are connected by a narrow tube with a tap which is closed. The pressure of air in one flask is double the other. After the tap is opened the common pressure in the flasks is 120.0 kPa. Find

a) the number of moles of gas used if volume of each flask is 5.6 m^3 at temperature 20°C and

b) the original pressure in the flasks.

(Ans.a) 552 mol. b)
80.0kPa, 160.0kPa)

calculations



Final pressure = 120.0 kPa

a) total number of moles

$$= \frac{2pV}{RT} = \frac{2(120 \times 10^3)5.6}{8.31(273 + 20)} = 552 \text{ mol}$$

b) conservation of mass
or number of moles

$$\frac{p_1(56)}{8.31(273 + 20)} + \frac{2p_1(56)}{8.31(273 + 20)} = 552$$

$$p_1 = 80.0 \text{ kPa}$$

$$p_2 = 160 \text{ kPa}$$

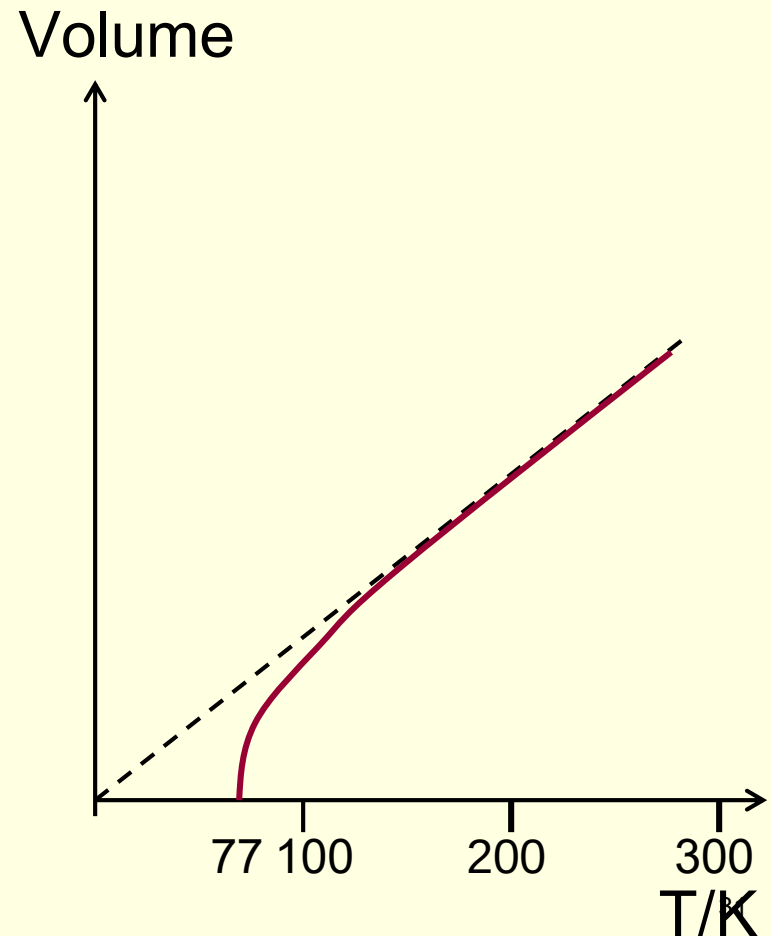
Real gases (info.)

- The ideal gas behaviour and the relationship between p , V and T are based on experimental observations of gases such as air, helium, nitrogen at **temperatures and pressures around room temperature**.
- In practice, if we change to more extreme conditions, such as **low temperatures and high pressures**, gases start to deviate from these laws as gas atoms exert significant intermolecular forces on each other.

Nitrogen (info.)

What happens when nitrogen is cooled down towards absolute zero?

- First it follows a good straight line at high temperature.
- As it approaches the temperature at which it condenses it deviates from ideal behaviour, and at 77 K it condenses to become liquid nitrogen.



The behaviour of real gases (info.)

- In our consideration of gases so far we have assumed that the intermolecular forces are zero and therefore that they follow the kinetic theory of gases exactly. However this is not the case with actual gases.

A gas that follows the gas laws precisely is known as an **ideal gas** and one which does not is called a **real gas**.

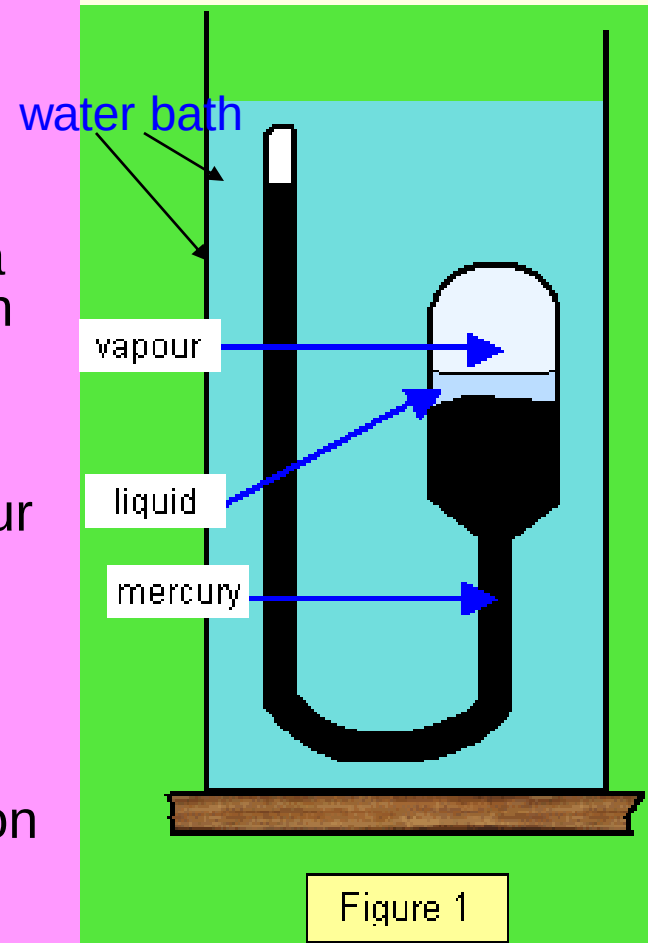
In 1847 Regnault constructed PV curves up to 400 atmospheres and found that Boyle's law was not obeyed at these high pressures.

Amagat went a stage further in 1892, working with nitrogen to pressures of some 3000 atmospheres (3×10^8 Pa) down a coal mine.

The behaviour of real gases (info.)

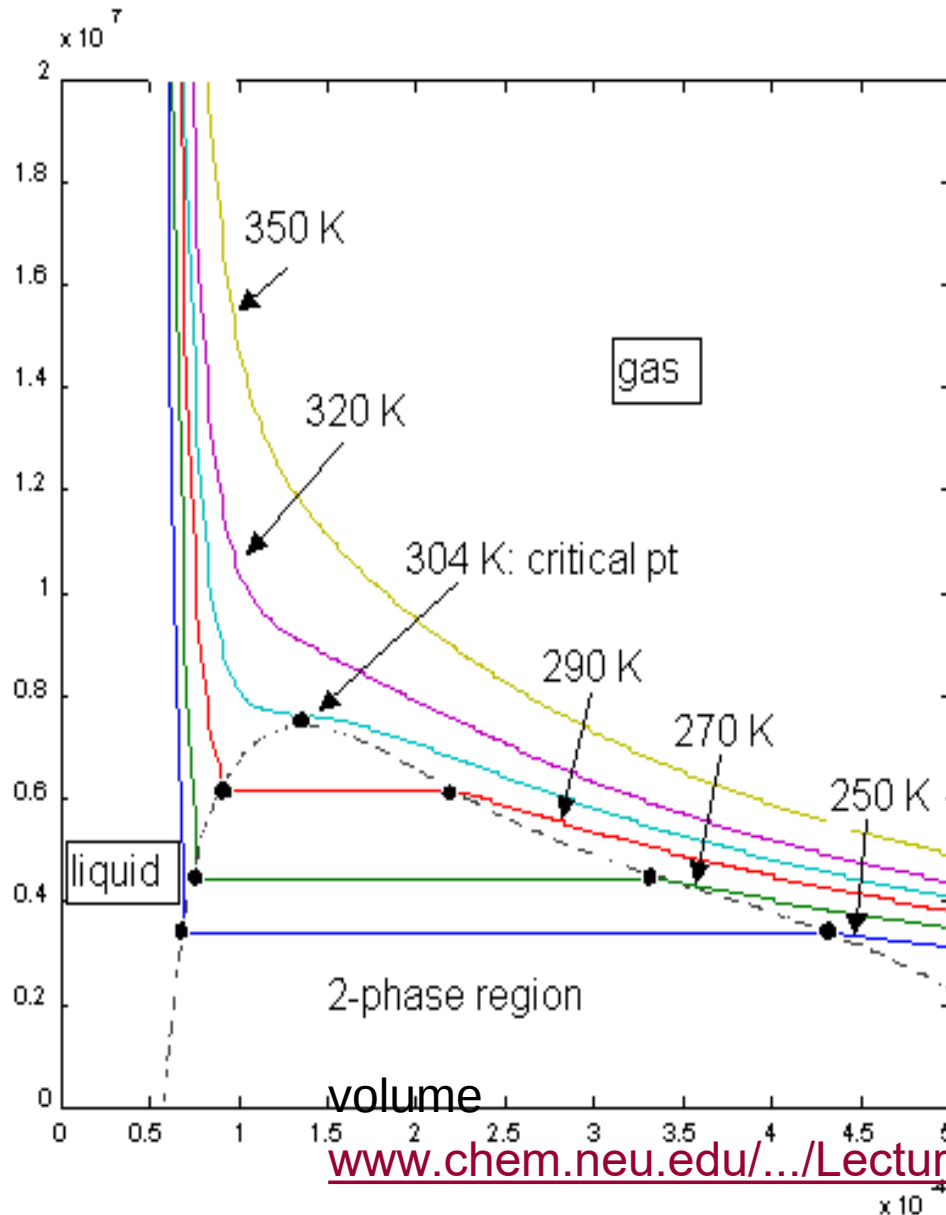
The idea that actual gases did not always obey the ideal gas equation was first tested by Cagniard de la Tour in 1822, using the apparatus shown in Figure 1.

A liquid such as water or ether was trapped in a tube and the end of the tube placed in a bath whose temperature could be controlled. The temperature was then varied and the behaviour of the liquid observed. The space above the liquid is obviously filled with vapour and it was noticed that at a particular temperature no difference could be seen between the liquid and vapour states - this was called the critical temperature. This phenomenon was not predicted by Boyle's law, which says nothing about the liquefaction of gases.



The temperature at which the liquid and vapour states can exist in equilibrium is called the critical temperature.

Real gases (info.)



- *Real gases liquefy.* As P is increased at constant T , at some point liquid will form. The liquification occurs at constant *pressure* (horizontal line on the P-V plot.)

Real gases **VERY RARELY BEHAVE LIKE IDEAL GASES** since

- There **IS** an **attraction between particle** (van der Waals forces)
- The **volume of particles** are **NOT** negligible, esp. at low temps & high-pressure since atoms/molecules are close together

HYDROGEN and **HELIUM** are the most **IDEAL** gases.

Also, Diatomic molecules and nonsymmetrical molecules & noble gases act the most ideal.

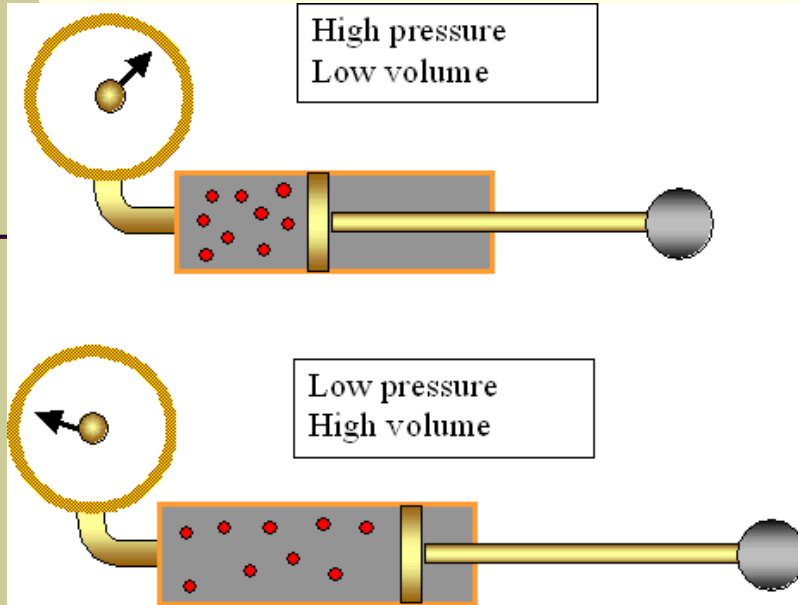
THE SMALLER THEY ARE THE MORE IDEAL THEY BEHAVE.

Summary

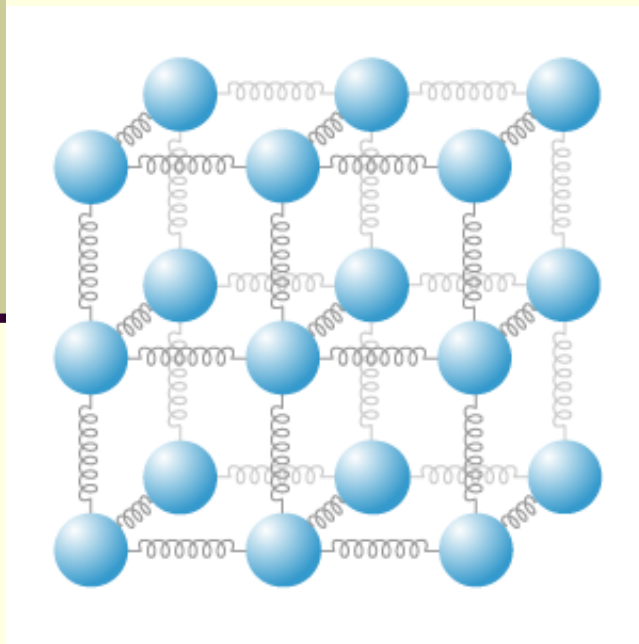
Summary

- Monatomic gases at low temperature and pressure are ideal gases.
- Ideal gases obey the gas laws; real gases do so to a first approximation.
- $pV = nRT$
- $L = 6.02 \times 10^{23} \text{ mol}^{-1}$

Intermolecular forces



In a solid, the molecules are bond together as if they are connected by springs. The molecules are in random vibration and the temperature of the solid is a measure of the average kinetic energy of the molecules.



Ludwig Boltzmann



was born in 1844 (Austria). Boltzmann was awarded a doctorate from the University of Vienna in 1866 for a thesis on the kinetic theory of gases supervised by Josef Stefan. After obtaining his doctorate, he became an assistant to his teacher Josef Stefan. Boltzmann taught at Graz, moved to Heidelberg and then to Berlin. In these places he studied under Bunsen, Kirchhoff and Helmholtz.

Attacks on his work continued and he began to feel that his life's work was about to collapse despite his defence of his theories. Depressed and in bad health, Boltzmann committed suicide just before experiment verified his work. On holiday with his wife and daughter at the Bay of Duino near Trieste, he hanged himself while his wife and daughter were swimming.

The ***Kinetic Theory of Matter*** is the statement of how we believe atoms and molecules, particularly in gas form, behave and how it relates to the ways we have to look at the things around us. The Kinetic Theory is a good way to relate the 'micro world' with the 'macro world.'

A statement of the Kinetic Theory is:

1. All matter is made of atoms, the smallest bit of each element. A particle of a gas could be an atom or a group of atoms.
2. Atoms have an energy of motion that we feel as temperature. The motion of atoms or molecules can be in the form of linear motion of translation, the vibration of atoms or molecules against one another or pulling against a bond, and the rotation of individual atoms or groups of atoms.

KINETIC THEORY OF MATTER (cont'd)

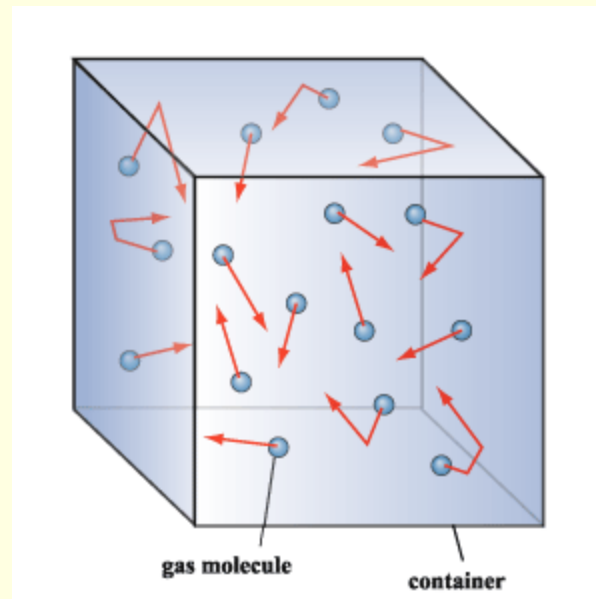
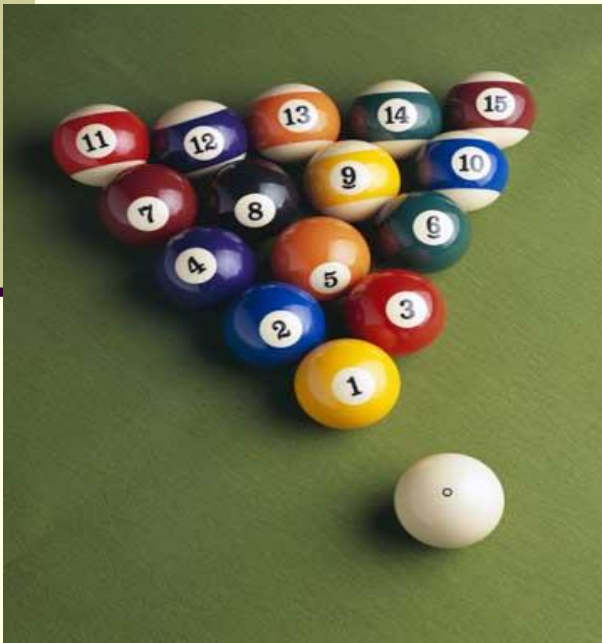
- 3) There is a temperature to which we can extrapolate, absolute zero, at which, theoretically, the motion of the atoms and molecules would stop.
- 4) The pressure of a gas is due to the motion of the atoms or molecules of gas striking the object bearing that pressure. Against the side of the container and other particles of the gas, the collisions are elastic (with no friction).
- 5) There is a very large distance between the particles of a gas compared to the size of the particles such that the size of the particle can be considered negligible.

Assumptions

- P** **Point molecules.** The volume of the molecules is negligible compared with the volume occupied by the gas, $V \gg b$
- I** **Intermolecular forces.** The molecules are far apart that the intermolecular forces are negligible.
- N** **Number.** There is a large number of molecules even in a small volume and that a large number of collisions occurs in a short time. The average of many impacts gives a smooth pressure.
- E** **Elastic collision.** Molecules are perfectly elastic sphere that they undergo elastic collisions.
- D** **Duration.** The duration of collision is negligible compared with the time between collision i.e. $t_2 \gg t_1$.

Brownian motion experiment

- Gives us evidence of continuous random motion of particles in liquids and gases.
- We can imagine that the particles as solid spherical tiny billiard balls.



Kinetic Theory

- When we study about ideal gas equation we are interested in **macroscopic properties** of gases (pressure, volume, and temperature that we can measured).
- It gives us a good description of gases in may different situation.
- It **does not explain** why gases behave in this way.
- Kinetic theory of gases is a theory which links these **microscopic properties** (mass, velocity, kinetic energy) of particles to the macroscopic properties of a gas.
- On the basis of these assumptions, it is possible to use Newtonian mechanics to show the gas laws,...gas particles move with a range of speeds.....

Temperature and kinetic energy

- Molecules in gases moved about randomly at high speed.
- They collide with one another and with the walls of their container.
- Collisions with the walls give rise to the pressure of the gas on the container.
- When a thermometer is placed in the container, the molecules collide with it and imparting their kinetic energy to the thermometer.
- At higher temperature, the molecules move faster or with greater kinetic energy. They give more kinetic energy to the bulb and the mercury rises higher.
- Hence the reading on the thermometer is an indication of the kinetic energy of the gas molecules

*Average kinetic energy
per molecule*

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

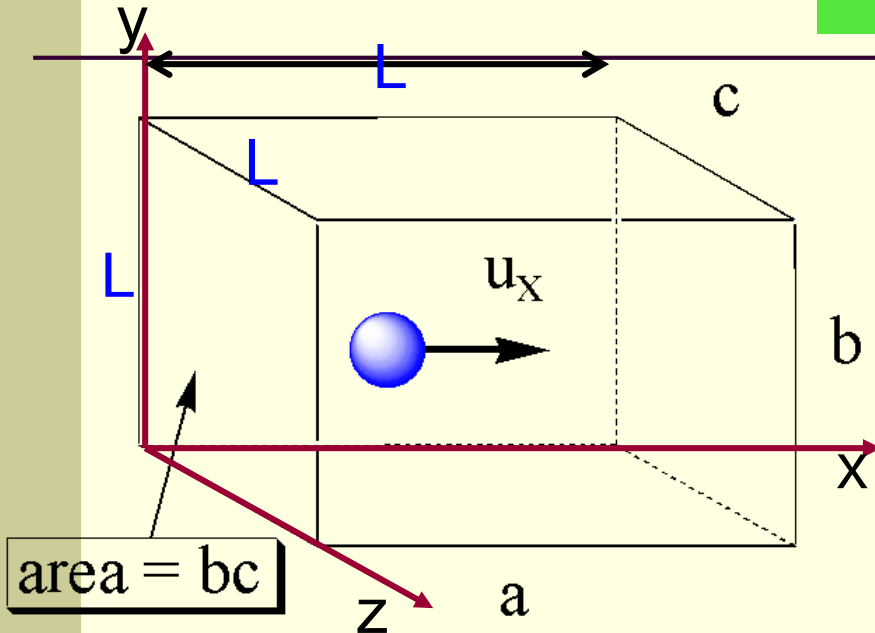
Kinetic theory and gas pressure

$$F = \frac{m(v - u)}{\Delta t}$$

- Kinetic theory states that the molecules of a gas **moves continuously** at random and often collides with the wall of the container.
- When a **molecule collides** with the wall of the container it undergoes change in momentum.
- The rate of change in momentum means that a **force acts** on the molecules. By Newton's third law of motion an equal but opposite force acts on the wall.
- Pressure is the average **force acting per unit area** as a result of impact of molecules of the gas on the wall of the container.

Derivation of

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$



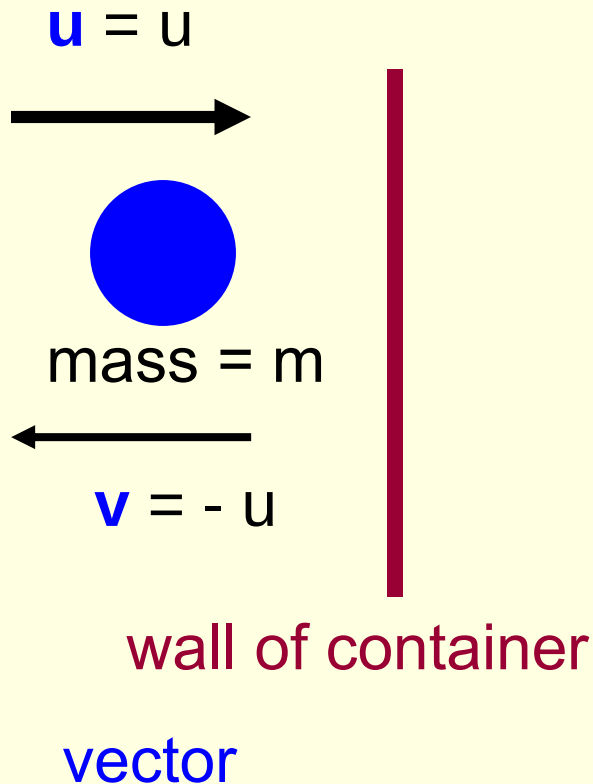
Consider a cubic container of sides L , containing N particles (monatomic) each of mass m .

Assumed

- all particles move in the **x direction** with the **same speed** u .
- particles are **monatomic**

Here we are interested in the particles colliding with the wall of the container, we are not interested in the collision between the particles

Change in momentum



For a molecule,

Change in momentum,

$$\Delta p = m (\mathbf{v} - \mathbf{u}) = -2mu$$

The time for the particle to impact the same face of the wall is

$$t = 2L/u$$

(as speed = dist./time)

Force on particle,

$$\begin{aligned} F &= \frac{\Delta p}{t} = - \frac{2mu}{2L / u} \\ &= - \frac{mu^2}{L} \end{aligned}$$

Pressure

Assumptions

- 1) All the molecules have the same velocity.
- 2) All molecules move in the x-direction

Force on wall by N molecules

$$F_T = NF = \frac{Nmu^2}{L}$$

Force on wall = - force on particle (NTLOM).

Pressure on wall

$$p = F_T/A = Nmu^2/L^3$$

$$p = Nmu^2/V \text{ where } V = L^3$$

Correcting for assumptions

- 1) in general the molecules can have any velocity in any direction,
- 2) $\frac{1}{3}$ of the molecules move in any of the three directions

$$pV = \frac{1}{3} Nm\langle c^2 \rangle$$

Pressure exerted by a gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

N = number of molecules

m = mass of a molecule

V = volume of container

c = speed of a molecule

$\langle c^2 \rangle$ = mean square speed

ρ = density of gas

Pressure of gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

depends on

- number of particles in the container, greater number of particles greater pressure.
- the greater the speed of gas the greater the pressure
- mass of gas and
- volume of container.

At higher temperature, the speed increase so pressure increases.

mean square speed, $\langle c^2 \rangle$

is the mean or average of the square of the speed of all the particles in the container.

- If there are n_1 particle with speed c_1 , n_2 particle with speed c_2 , n_3 with speed c_3 and n_n particles with speed c_n , then
- Total number of particles

$$N = n_1 + n_2 + \dots n_n$$

$$\langle c^2 \rangle = \frac{n_1 c_1^2 + n_2 c_2^2 + n_3 c_3^2 + \dots n_n c_n^2}{N}$$

Root mean square speed (c_{rms})

$$c_{rms} = \sqrt{\langle c^2 \rangle} = \sqrt{\frac{3kT}{m}}$$

- It is the square root of mean square speed.
- c_{rms} is directly proportional to the square root of absolute temperature..
- c_{rms} is inversely proportional to $\sqrt{m}$. For a mixture of gases in a container at thermal equilibrium, the heavier gas has a smaller root mean square speed.

Example 25.1

Five molecules have speeds 100, 200, 300, 400 and 500 m/s. Find

- a) their mean speed,
- b) mean square speed, and
- c) root mean square speed.

(Ans. a) 300 m/s; b) $1.1 \times 10^5 \text{ m}^2\text{s}^{-2}$; c) 330 m/s)

Solution

There are 5 molecules

a) Total speed, c_T
$$= (100 + 400 + 200 + 300 + 500) = 3(500)$$
$$= 1500$$
$$\langle c \rangle = 1500/5$$
$$= 300 \text{ m/s}$$

b) $\langle c^2 \rangle = (100^2 + 200^2 + 300^2 + 400^2 + 500^2)/5$
$$= 1.1 \times 10^5 \text{ m}^2\text{s}^{-2}$$

c) $c_{\text{rms}} = \sqrt{\langle c^2 \rangle} = 330 \text{ m/s}$

Example 25.2

The density of air at s.t.p. is 1.3 kg m^{-3} and the atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$.

Calculate

- a) the means square speed, and
- b) the root mean square speed.

(Ans.: a) $2.33 \times 10^5 \text{ m}^2\text{s}^{-2}$; b) 482 m/s)

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

$$\text{a) } 1.01 \times 10^5 = \frac{1}{3} (1.3) \langle c^2 \rangle$$

$$\langle c^2 \rangle = 2.33 \times 10^5 \text{ m}^2\text{s}^{-2}$$

$$\begin{aligned} \text{b) } c_{\text{rms}} &= \sqrt{\langle c^2 \rangle} \\ &= 482 \text{ m/s} \end{aligned}$$

Average translational kinetic energy of a molecule

From kinetic theory of gases,

$$pV = \frac{1}{3} Nm\langle c^2 \rangle$$

Ideal gas equation

$$pV = NkT$$

Since both equations are for ideal gas

$$\frac{1}{3} Nm\langle c^2 \rangle = NkT$$

$$m\langle c^2 \rangle = 3kT$$

Average kinetic energy of a molecule

$$\begin{aligned} E_k &= \frac{1}{2} m\langle c^2 \rangle \\ &= \frac{3}{2} kT \end{aligned}$$

■ this equation shows that the mean kinetic energy is directly proportional to the thermodynamic temperature.

Internal Energy of ideal gas

Total kinetic energy

$$E_{kT} = \frac{1}{2} mN\langle c^2 \rangle$$
$$= \frac{3}{2} NkT \text{ or } \frac{3}{2} nRT$$

- for ideal gas there is no intermolecular force between the particles.
- the energy is totally kinetic energy.

Internal energy

$$U = \text{total kinetic energy}$$
$$= \frac{3}{2} NkT \text{ or } \frac{3}{2} nRT$$

(internal energy is mainly kinetic energy)

- an increase in temperature of the gas means an increase in total kinetic energy of the gas, thus an increase in internal energy

Internal Energy of ideal gas

$$U = \frac{3}{2} NkT \text{ or } \frac{3}{2} nRT$$

- The **internal energy** of an ideal monatomic gas is directly proportional to its absolute temperature.
- This is true regardless of the molecular structure of the gas. However, the expression for U will be a bit different for gases that are not monatomic.
- **Absolute temperature** of a gas is directly proportional to its average random kinetic energy per molecule.
- This means that if the **absolute temperature** of a gas is doubled by heat transfer, for example, from 200 K to 400 K, then its **internal energy** is also doubled.
- This does not apply to the **Celsius temperature**, since its zero points are not referenced to the zero-point energy.

Mass, kinetic energy and temperature.

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

- KE is directly proportional to the absolute temperature, T. As the temperature is double the average KE per molecule increases.
- Air is a mixture of several gases for example nitrogen, oxygen and carbon dioxide.

- In a sample of **air**, the mean KE of the nitrogen molecules is the same as that of oxygen and carbon dioxide molecules.
- Carbon dioxide molecules have a **greater mass** than oxygen molecules.
- Since the kinetic energy is the same, oxygen molecules move **faster** than carbon dioxide molecules.

Self Test 11

- 1) What is an ideal gas?
- 2) The pressure p in an ideal gas is given by the expression

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

State the meaning of each of the symbols in the equation.

- 3) State the equation of state of an ideal gas and the meaning of the symbols used.

- 1) A gas that obeys Boyle's law, the gas laws and the equation of state for all temperature, pressure and volume.
- 2) N = number of molecules, m =mass of particles, V =volume of container and $\langle c^2 \rangle$ is the mean square speed.
- 3) $pV=nRT$
 p = pressure of gas
 V = volume of container
 n = number of mole of gas
 R = molar gas constant
 T = absolute temperature

4) State the basic assumptions of the kinetic theory of gases.

5) State the meaning of each of the symbols in the equation.

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

What is the significant of $\frac{1}{2} m \langle c^2 \rangle$?

6) Can we say that since yesterday the temperature was 10°C and today the temperature is 20°C, then today is twice as hot?

5) m = mass of particle, $\langle c^2 \rangle$ = mean square speed; k = Boltzmann constant; T = absolute temperature.

It is the average kinetic energy of a particles in a gas.

6) No, as the kinetic energy is not double. Kinetic energy is proportional to absolute temperature rather than Celsius temperature.

PYP 11.1

A kinetic theory formula relating the pressure p and the volume V of a gas to the root-mean-square speed of its molecules is

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

Ans. A

In this formula, what does the product Nm represent?

- A. the mass of gas present in the volume V .
- B. the number of molecules in unit volume of the gas
- C. the total number of molecules in one mole of gas
- D. the total number of molecules present in volume V

PYP 11.2

The simple kinetic theory of gases may be used to derive the expression relating the pressure p to the density ρ of gas.

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

In this expression, what does $\langle c^2 \rangle$ represent?

A. the average of the squares of the speeds of the gas molecules

B. the root-mean-square speed of the gas molecule $c_{rms} = \sqrt{\langle c^2 \rangle}$

C. the square of the average speed of the gas molecules $\langle c \rangle^2$

D. the sum of the squares of the speeds of the gas molecules. $c_1^2 + c_2^2 + c_3^2 + \dots c_n^2$

Ans. A

PYP 11.3

*Average kinetic energy
per molecule*

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

The molecules of an ideal gas at thermodynamics (absolute) temperature T have a root-mean-square speed c_r . The gas is heated to temperature $2T$. What is the new root-mean-square speed of the molecules?

Solution

m and k are constant

Let the new root-mean-square speed be x

$$\langle c^2 \rangle / T = \text{constant}$$

$$c_r^2 / T = x^2 / 2T$$

$$x = (\sqrt{2}) c_r$$

$$\frac{c_1^2}{T_1} = \frac{c_2^2}{T_2}$$

PYP 11.4

The pressure p of a gas occupying a volume V and containing N molecules of mass m and mean square speed $\langle c^2 \rangle$ is given by

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

The density of argon at a pressure 1.00×10^5 Pa and at a temperature 200 K is 1.60 kg m^{-3} . What is the root mean square speed of argon molecules at this temperature?

Solution

Note: $\rho = Nm/V$

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

$$1.00 \times 10^5 = \frac{1}{3} 1.6 \langle c^2 \rangle$$

$$c_r = 433 \text{ m/s}$$

PYP 11.5

$$pV = nRT$$

An ideal gas has volume
0.50 m³ at a pressure
1.01x10⁵ Pa and
temperature 17°C.

b) Calculate, for the gas, the
number of

i) moles,
number =

ii) molecules.
number =

Solution

$$\begin{aligned} \text{bi) } pV &= nRT \\ 1.01 \times 10^5 (0.5) &= n(8.31)(273 + 17) \\ n &= 21 \text{ moles} \end{aligned}$$

$$\begin{aligned} \text{ii) } n &= N/N_A \\ 21 &= N / 6.02 \times 10^{23} \\ N &= 1.26 \times 10^{25} \end{aligned}$$

PYP 11.5

c) Each molecule may be considered to be sphere of radius $1.2 \times 10^{-10} \text{m}$. Calculate

i) the volume of one molecule of the gas,
volume =

ii) the volume of all the molecules.
volume =

Solution

ci) volume of one molecule
$$= \left(\frac{4}{3}\right)\pi r^3$$
$$= \left(\frac{4}{3}\right)\pi (1.2 \times 10^{-10})^3$$
$$= 7.24 \times 10^{-30} \text{ m}^3$$

ii) volume
$$= 7.24 \times 10^{-30} (1.26 \times 10^{25})$$
$$= 9.12 \times 10^{-5} \text{ m}^3$$

PYP 11.5

- di) State the assumption made in the kinetic theory of gases for the volume of the molecules of an ideal gas.
- dii) Comment on your answer to cii) with reference to this assumption.

Solution

di) The volume of the molecules is negligible when compare to the volume of the container.

dii) compare

$$\frac{\text{volume of container}}{\text{volume of molecules}} = \frac{0.5}{9.12 \times 10^{-5}} \\ = 5.5 \times 10^3$$

volume of container is very much greater than volume of molecules.

PYP 11.5

- ai) The kinetic theory of gases leads to the equation

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

Explain the significance of the quantity $\frac{1}{2} m \langle c^2 \rangle$

- ii) Use the equation to suggest what is meant by the absolute zero of temperature. [3]

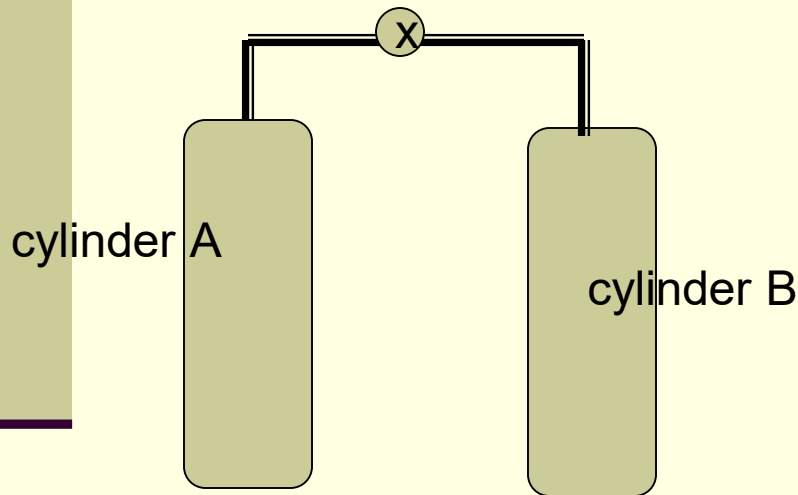
Solution

- ai) It is the average kinetic energy of a molecule.

- ii) At absolute zero of temperature i.e. $T = 0$, the kinetic energy is zero, i.e. the molecules are at rest.

PYP 11.5

- b) Two insulated gas cylinders A and B are connected by a tube of negligible volume, as shown in Fig.



Each cylinder has an internal volume of $2.0 \times 10^{-2} \text{ m}^3$.

Initially, the tap is closed and cylinder A contains 1.2 mol. of an ideal gas at temperature of 37°C . Cylinder B contains the same ideal gas at pressure $1.2 \times 10^5 \text{ Pa}$ and a temperature.

- i) Calculate the amount, in mole of the gas in cylinder B.

Solution

$$\begin{aligned} \text{i) } pV &= nRT \\ 1.2 \times 10^5 (2.0 \times 10^{-2}) &= \\ n(8.31)(273 + 37) &= \\ n &= 0.932 \text{ mol.} \end{aligned}$$

PYP 11.5

$$pV = nRT$$

bii) The tap is opened and some gas flows from cylinder A to cylinder B. Using the fact that the total amount of gas is constant, determine the final pressure of the gas in the cylinders.

For two
containers of
equal volume

Solution

bii) Let the final pressure in each container be p .
total amount initially
= total amount finally

$$1.2 + 0.93 = n_A + n_B$$

$$2 \frac{p(2 \times 10^{-2})}{8.31(273 + 37)}$$

$$p = 1.37 \times 10^5 \text{ Pa}$$

Boltzmann
constant = R/N_A

$$pV = NkT$$

a gas that obeys
gas laws, $PV=nRT$
at all T , p and V

Absolute
temperature
 $T = \theta + 273.15$

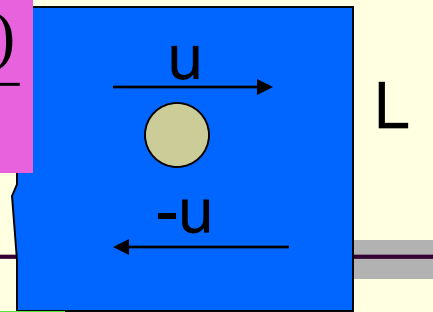
Ideal gas

N = number of molecules
 m = mass of a molecule
 V = volume of container
 c = speed of a molecule
 $\langle c^2 \rangle$ = mean square speed
 $\rho = Nm/V$ = density of gas

$$F = \frac{m(v-u)}{\Delta t}$$

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$



kinetic theory

Assumptions

Point molecules
Elastic collision
Large Number
Duration of
collision
No intermolecular
forces